

Automation Testing & Error Handling Report

Completed validation cases for scripts, data quality, and safe operational behavior

Test	Expected behavior	Result
Valid IP in approved CIDR	Retained in allow-list	PASS
Duplicate valid IP	One retained; duplicate recorded	PASS
Valid IP outside approved CIDR	Removed and logged unauthorized	PASS
Malformed IP	Removed and logged invalid	PASS
Dry-run mode	Summary/log generated; output list not changed	Supported by CLI design
Missing script argument	argparse rejects execution with usage message	Supported by CLI design
CSV repeated failures	Medium alert at threshold	PASS in generated sample
Failures then success	High alert generated	PASS in generated sample
SQL repeated failures by source	Source aggregates returned	PASS in SQLite output
Bash optional tool absent	Human-readable message / guarded command	Supported by script design

Test Execution Evidence

PASS: allowlist evaluation Spreadsheet runtime warmup failed during python startup Traceback (most recent call last): File "/tmp/tmp.L2TH2Y5coc/artifact_tool_v2-2.8.22/artifact_tool/patches/warm_spreadsheet_runtime_on_startup.py", line 26, in warm_spreadsheet_runtime_on_startup File "/tmp/tmp.L2TH2Y5coc/artifact_tool_v2-2.8.22/artifact_tool/spreadsheet_warmup.py", line 772, in warm_spreadsheet_runtime File "/tmp/tmp.L2TH2Y5coc/artifact_tool_v2-2.8.22/artifact_tool/rpc/connection.py", line 37, in get_or_create_client File "/tmp/tmp.L2TH2Y5coc/artifact_tool_v2-2.8.22/artifact_tool/rpc/daemon.py", line 124, in start_daemon TimeoutError: Timed out waiting for artifact tool daemon socket. Set ARTIFACT_TOOL_RPC_DAEMON_STARTUP_TIMEOUT_S= to increase the limit.

Error-Handling Design

- Python CLI arguments are required so the script does not silently operate on the wrong file.
- Invalid IP values are recorded rather than causing the entire allow-list job to fail.
- Unauthorized addresses are separated from syntactically invalid addresses for clearer review.
- Bash commands that may not exist on every distribution are checked or guarded.
- Automation outputs are separated from input data to make review and rollback easier.